

一、精确问题陈述

目标是严格解释 printed page 1104 的 equation (5.5) 是否能由此前内容推出。给定

$$\rho_{X_0, t_0}(F, t) = \frac{1}{4\pi(t_0 - t)} \exp\left(-\frac{|F - X_0|^2}{4(t_0 - t)}\right),$$

以及固定原坐标 cutoff $\phi \in C_0^\infty(B_{2r}(X_0))$ 、 $\phi \equiv 1$ on $B_r(X_0)$ 时的 localized Gaussian quantity

$$\Psi(X_0, t_0; t) = \int_{\Sigma_t} (\cos \alpha)^{-p} \phi(F) \rho_{X_0, t_0}(F, t) d\mu_t.$$

Theorem 3.3 给出一侧 differential inequality:

$$\frac{d}{dt} \left(e^{c_1 \sqrt{t_0 - t}} \Psi(X_0, t_0; t) \right) \leq -\text{nonnegative square terms} + c_2 e^{c_1 \sqrt{t_0 - t}}.$$

在 Section 5 的 parabolic rescaling 中, 令

$$\begin{aligned} F_\lambda(x, s) &= \lambda(F(x, T + \lambda^{-2}s) - X_0), \quad s < 0, \\ d\mu_s^\lambda &= \lambda^2 d\mu_{T + \lambda^{-2}s}, \quad w_\lambda = (\cos \alpha_\lambda)^{-p}. \end{aligned}$$

printed equation (5.5) 声称: 对 fixed $-1 < s_1 < s_2 < 0$,

$$G_\lambda(s_2) - G_\lambda(s_1) \rightarrow 0 \quad \text{as } \lambda \rightarrow \infty,$$

其中字面显示的 endpoint quantity 是

$$G_\lambda(s) = e^{c_1 \sqrt{T - (T + \lambda^{-2}s)}} \int_{\Sigma_s^\lambda} w_\lambda \phi_R(F_\lambda) (-s)^{-1} \exp\left(-\frac{|F_\lambda|^2}{4(-s)}\right) d\mu_s^\lambda.$$

原始请求是: 严格说明 gradient flow 中 printed page 1104 的 equation (5.5) 如何由前文推出; 若不能按原文推出, 则指出 exact gap, 并给出 corrected claim。后续目标进一步要求: 给出 remaining moving-cutoff lemma 的完整证明, 或者严格证明该 lemma 为 false / underived, 并给出实际成立的 strongest corrected theorem。

二、主要数学进展

Proven: 固定原坐标 cutoff 版本严格成立。

令 $\chi \in C_0^\infty(B_{2r}(X_0))$ 是一个固定 admissible cutoff, $\chi = 1$ on $B_r(X_0)$ 。定义

$$\Psi_\chi(t) = \int_{\Sigma_t} (\cos \alpha)^{-p} \chi(F) \frac{1}{4\pi(T - t)} \exp\left(-\frac{|F - X_0|^2}{4(T - t)}\right) d\mu_t,$$

以及

$$Q_\chi(t) = e^{c_1 \sqrt{T - t}} \Psi_\chi(t).$$

由 differential inequality 舍去 nonpositive square terms, 得到

$$Q'_\chi(t) \leq c_2 e^{c_1 \sqrt{T-t}}.$$

因为右端在 $[a, T)$ 上 integrable, 设

$$A(t) = \int_t^T c_2 e^{c_1 \sqrt{T-u}} du.$$

则 $A(t) \rightarrow 0$, 且

$$(Q_\chi + A)'(t) = Q'_\chi(t) - c_2 e^{c_1 \sqrt{T-t}} \leq 0.$$

所以 $Q_\chi + A$ monotone decreasing 且 bounded below, 由此 $Q_\chi(t)$ 有 finite one-sided limit:

$$\lim_{t \uparrow T} Q_\chi(t) = L_\chi < \infty.$$

接着作 exact rescaling。令

$$t_\lambda(s) = T + \lambda^{-2}s.$$

则

$$T - t_\lambda(s) = \lambda^{-2}(-s), \quad F - X_0 = \lambda^{-1}F_\lambda, \quad d\mu_s^\lambda = \lambda^2 d\mu_{t_\lambda(s)}.$$

Kähler angle 在 positive homothety 下 invariant, 所以

$$(\cos \alpha_\lambda)^{-p} = (\cos \alpha)^{-p}.$$

于是

$$\begin{aligned} & \int_{\Sigma_s^\lambda} (\cos \alpha_\lambda)^{-p} \chi(X_0 + \lambda^{-1}F_\lambda)(-s)^{-1} \exp\left(-\frac{|F_\lambda|^2}{4(-s)}\right) d\mu_s^\lambda \\ &= 4\pi \Psi_\chi(t_\lambda(s)). \end{aligned}$$

乘上 exponential factor 得

$$J_\lambda^\chi(s) = 4\pi Q_\chi(t_\lambda(s)).$$

因此对 fixed $s_1 < s_2 < 0$,

$$J_\lambda^\chi(s_2) - J_\lambda^\chi(s_1) = 4\pi(Q_\chi(t_\lambda(s_2)) - Q_\chi(t_\lambda(s_1))) \rightarrow 0,$$

因为两个时间都趋向同一个 terminal limit L_χ 。

这里 4π 只是 normalization: 若使用 normalized kernel $[4\pi(-s)]^{-1}$, 则没有这个 factor; 若使用 unnormalized $(-s)^{-1}$, 两端共同乘以 4π , 不影响 zero limit。

Proven: printed fixed scaled cutoff 与 fixed original cutoff 不是同一对象。

printed expression 使用 $\phi_R(F_\lambda)$, 其中 ϕ_R fixed in scaled coordinates。换回 original coordinates:

$$\phi_R(F_\lambda) = \phi_R(\lambda(F - X_0)) =: \chi_{R,\lambda}(F).$$

也就是说

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

该 cutoff 支撑在 $B_{2R/\lambda}(X_0)$, 并在 $B_{R/\lambda}(X_0)$ 上等于 1, 所以它随 λ 改变。

因此 printed endpoint quantity 精确等于

$$4\pi$$

乘以一个 Section 3 type quantity, 但这个 quantity 的 cutoff 是 $\chi_{R,\lambda}$, 不是一个 fixed χ 。固定函数 $Q_\chi(t)$ 的 terminal-limit argument 不能直接应用到 moving family $Q_{\chi_{R,\lambda}}(t)$ 。

Proven: individual terminal limits for moving families 不足以推出 two-time limit.

存在抽象反例:

$$Q_n(t) = n^2(-t), \quad t \in (-1, 0).$$

每个 $Q_n \geq 0$, locally absolutely continuous, 且

$$Q'_n(t) = -n^2 \leq 0.$$

每个 Q_n 都有 terminal limit:

$$\lim_{t \uparrow 0} Q_n(t) = 0.$$

但对 fixed $s_1 < s_2 < 0$,

$$Q_n(n^{-2}s_2) - Q_n(n^{-2}s_1) = s_1 - s_2 \neq 0.$$

所以“每个 moving cutoff quantity 各自有 terminal limit”并不 imply diagonal parabolic-scale two-time convergence。

Proven: literal fixed scaled compact cutoff 在 static plane model 中失败。

在 flat symplectic plane $P = \mathbb{R}^2 \times \{0\} \subset \mathbb{C}^2$ 上取 stationary immersion $F(x, t) = x$ 。此时

$$\cos \alpha = 1, \quad H = 0, \quad V = 0,$$

parabolic rescaling 后仍是同一 plane, 且 weight 为 1。

取 nontrivial radial compact cutoff ϕ_R , 满足

$$\phi_R = 1 \text{ on } B_R, \quad \phi_R = 0 \text{ outside } B_{2R},$$

并且 radial nonincreasing。定义

$$I(s) = \int_P \phi_R(Y)(-s)^{-1} \exp\left(-\frac{|Y|^2}{4(-s)}\right) dA(Y).$$

uncut integral 恒等于

$$\int_P (-s)^{-1} \exp\left(-\frac{|Y|^2}{4(-s)}\right) dA(Y) = 4\pi.$$

但 compactly cut integral 不恒定。令 $s_1 = -1$, $s_2 = -1/4$, 换元 $Y = \sqrt{a}Z$, $a = -s$, 得

$$I(-a) = \int_P \phi_R(\sqrt{a}Z) e^{-|Z|^2/4} dA(Z).$$

因为 $a_2 = 1/4 < a_1 = 1$ 且 ϕ_R radial nonincreasing,

$$\phi_R(\sqrt{a_2}Z) \geq \phi_R(\sqrt{a_1}Z),$$

并在 cutoff annulus 的正面积子集上严格大。因此

$$I(-1/4) > I(-1).$$

在 static plane model 中

$$G_\lambda(s) = e^{c_1 \lambda^{-1} \sqrt{-s}} I(s),$$

所以

$$\lim_{\lambda \rightarrow \infty} (G_\lambda(-1/4) - G_\lambda(-1)) = I(-1/4) - I(-1) > 0.$$

这 disproves standalone fixed-scaled-cutoff lemma。

但该 plane 不是 full Section 5 compact first-singular-flow counterexample: 它 noncompact, 没有 finite first singular time, 且 X_0 在 T 时不是 singular point。

Proven: moving cutoff derivative terms 是 order-one annular flux。

对 literal cutoff

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)),$$

有

$$D\chi_{R,\lambda} = \lambda D\phi_R, \quad D^2\chi_{R,\lambda} = \lambda^2 D^2\phi_R.$$

把它放入 pre-(5.5) monotonicity calculation 后, cutoff-gradient、cutoff-Laplacian、heat-kernel-gradient terms 在 parabolic change of variables $t = T + \lambda^{-2}s$ 下变为 fixed scaled annulus

$$R < |F_\lambda| < 2R$$

上的 spacetime integrals:

$$\int_{s_1}^{s_2} \int_{\Sigma_s^\lambda} w_\lambda \rho_s(F_\lambda) [D\phi_R(F_\lambda) \cdot v_\lambda + \Delta_{\Sigma_s^\lambda}(\phi_R \circ F_\lambda) + 2\langle \nabla_{\Sigma_s^\lambda} \log \rho_s, \nabla_{\Sigma_s^\lambda}(\phi_R \circ F_\lambda) \rangle] d\mu_s^\lambda ds.$$

这里没有 λ^{-1} 或 λ^{-2} prefactor。也就是说, moving cutoff error 并不会因 scaling 自动变成 $o(1)$ 。要证明 literal moving-cutoff endpoint limit, 必须额外证明 annular flux vanishing / cancellation, 或某种 uniform moving-cutoff convergence-to-terminal-limit。

Conditional: 若 endpoint weighted measures 两时刻趋于同一 nonzero cone/plane, 则 literal compact cutoff 反而有 positive defect。

若在 s_1, s_2 两个 endpoint 上, pushed-forward weighted measures 弱收敛到同一个 nonzero two-dimensional cone measure ν_{cone} , 则对 suitable radial compact cutoff,

$$\int \phi_R(Y)(-s_2)^{-1} e^{-|Y|^2/[4(-s_2)]} d\nu_{\text{cone}} > \int \phi_R(Y)(-s_1)^{-1} e^{-|Y|^2/[4(-s_1)]} d\nu_{\text{cone}}.$$

因此 literal compact scaled cutoff endpoint difference 的 limit 是 strictly positive, 而不是 zero。这个是 conditional obstruction, 不是 full counterexample, 因为 same-limit conical convergence 不能在 (5.5) 前 noncircularly 使用。

三、主要障碍

核心障碍只有一个: printed $\phi_R(F_\lambda)$ 是 fixed in scaled coordinates, 但它在 original coordinates 中是 moving cutoff

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

Theorem 3.3 的 terminal-limit argument 控制的是一个 fixed localized Gaussian quantity。它给出的是

$$Q_\chi(t) \rightarrow L_\chi,$$

不是 moving family $Q_{\chi_{R,\lambda}}(t)$ 在 parabolic-scale times 上 uniformly 接近各自 terminal limit。

把 moving cutoff 直接代回 monotonicity calculation 也不能解决问题, 因为 cutoff derivative terms 在 scaled variables 中变成 order-one annular flux, 没有 small parameter, 也没有 sign。static plane computation 说明该 flux 不是形式误差: 即使 velocity、curvature、flow-energy 都为零, compact cutoff 仍能看到 Gaussian width 随 s 改变而产生的非零 endpoint difference。

因此，标准 fixed-cutoff monotonicity、one-sided limit、local area bound、fixed-time Radon compactness 都不足以推出 literal (5.5)。需要一个新的 noncircular bridge。

四、方法时间线

stage	question addressed	conclusion established	effect on the approach
fixed-cutoff terminal limit	(3.6) 的 one-sided inequality 是否给出 terminal limit	对 fixed cutoff χ , $Q_\chi(t)$ has finite one-sided limit as $t \uparrow T$	fixed-cutoff endpoint route is valid
exact rescaling	rescaled endpoint quantity 与 Section 3 quantity 如何对应	unnormalized scaled integral equals $4\pi\Psi_\chi$ for fixed pullback cutoff $\chi(X_0 + \lambda^{-1}F_\lambda)$	4π is harmless normalization
cutoff comparison	printed $\phi_R(F_\lambda)$ 是否等于 fixed-cutoff pullback	$\phi_R(F_\lambda)$ corresponds to moving cutoff $\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0))$	identifies the real gap
formal moving-family test	individual terminal limits 是否 imply diagonal two-time convergence	false, even for monotone nonnegative scalar families	fixed-cutoff argument cannot be applied familywise
static plane model	fixed scaled compact cutoff 是否 intrinsically harmless	false in standalone model; endpoint difference can be strictly positive	shows the obstruction is real
annular flux audit	pre-(5.5) estimates 是否 hide a uniform moving-cutoff estimate	no; derivative terms become order-one annular flux	literal moving-cutoff lemma remains unproved
compactness audit	fixed-time local area compactness 是否 repairs the gap	only gives separate fixed-time weak limits, not same-limit or flux control	insufficient for literal endpoint limit
corrected theorem	what unconditional statement is actually proved	fixed-original-cutoff endpoint theorem with $\chi(X_0 + \lambda^{-1}F_\lambda)$	valid replacement for rigorous use

五、当前状态与下一步

当前问题在 literal fixed compact scaled cutoff reading 下尚未解决：没有 verified proof，也没有 full compact first-singular-flow counterexample。已经 proved 的是 stronger diagnosis：原文从 (3.6) 到 printed (5.5) 的 common-limit step 只严格证明 fixed-original-cutoff replacement；literal $\phi_R(F_\lambda)$ 版本需要一个额外 noncircular moving-cutoff theorem。

最强 unconditional corrected statement 是： $\boxed{\begin{minipage}{0.92\linewidth} \text{Let } (X_0) \text{ be the blow-up point and } (T) \text{ the first singular time. Let } (\chi \in C_0^\infty(B_{2r}(X_0))) \text{ be fixed with } (\chi=1) \text{ on } (B_r(X_0)). \text{ For fixed } (-1 < s_1 < s_2 < 0), \text{ define } \text{Then } \text{The same statement with normalized kernel } ([4\pi(-s)]^{-1}) \text{ is equivalent. } \end{minipage}}$

剩余 literal moving-cutoff lemma 若要成立，必须另证： $\boxed{\begin{minipage}{0.92\linewidth} \text{For the moving cutoff } \text{let } \text{If } (L_{\{R,\lambda\}} = \lim_{t \uparrow T} Q_{\{R,\lambda\}}(t)), \text{ prove noncircularly that for each } (i=1,2), \text{ } \text{Equivalently, prove annular cutoff-flux cancellation/vanishing on } \text{at the parabolic scale } (T-t \sim \lambda^{-2}). \end{minipage}}$

在这个 bridge 被 proved 或 full-hypothesis counterexample 被 constructed 之前，不能把 printed literal compact-scaled-cutoff equation (5.5) 当作已严格推出。